

AI-Powered Smart Forest Fire Detection and Environmental Monitoring System

CSA1013 – SOFTWARE ENGINEERING

Submitted by
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PROBLEM STATEMENT:

Scenario-Based Requirement Analysis Software Requirement Specification (SRS)

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.
Total Maximum Marks	20 Marks			

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements, actors, use cases, data model, interfaces, and system features of the AI-Powered Smart Forest Fire Detection and Environmental Monitoring System. It translates the problem statement and industry context supplied for this assignment into a scenario-based requirement analysis that a development team, Product Owner, and quality-assurance function can use as the single source of truth for design, implementation, and acceptance testing.

The document follows the structure prescribed for this exercise: scenario analysis, functional and non-functional requirement identification, actor and use-case modelling (including a use-case diagram), the SRS body itself, assumptions and constraints, and a closing validation pass that checks the specification for completeness, ambiguity, redundancy, and inconsistency.

1.2 Document Conventions

Functional requirements are identified with the prefix FR- followed by a module number and sequence (e.g., FR-3.2). Non-functional requirements use the prefix NFR-. Use cases use the prefix UC-. Priority is expressed using MoSCoW (Must, Should, Could, Won't) so that this SRS remains directly traceable to the sprint-planning and backlog-prioritisation artefacts produced for this project.

1.3 Intended Audience

- Product Owner / Forest Department policy representatives, to approve scope and priority.
- The Scrum development team (backend, frontend, ML, and QA engineers), as the primary implementers.
- Forest Fire Management Authorities and Emergency Response Services, as key business stakeholders and reviewers of detection/alerting behaviour.
- System Administrators, as owners of the non-functional security and access-control requirements.
- Academic evaluators assessing this document as a Software Engineering coursework deliverable.

1.4 Project Context

Forest fires cause severe ecological damage, biodiversity loss, and economic impact, and climate change has increased both the frequency and intensity of wildfires. Conventional detection methods — manual observation and low-frequency satellite passes — typically confirm a fire only after it has already spread, by which point containment is far more difficult and damage is far more extensive. This project addresses that gap by fusing IoT ground sensing, AI-based computer vision, drone and satellite imagery, and weather/drought data into a single continuous monitoring and early-warning platform.

1.5 Definitions, Acronyms, and Abbreviations

Term	Definition
IoT	Internet of Things — networked ground sensors reporting temperature, humidity, smoke, and wind.
AI / CV	Artificial Intelligence / Computer Vision — models used for fire-signature detection in imagery and sensor data.

Term	Definition
RBAC	Role-Based Access Control — access rights assigned according to a user's organisational role.
SRS	Software Requirement Specification — this document.
FR / NFR	Functional Requirement / Non-Functional Requirement.
UC	Use Case.
MQTT	Lightweight publish-subscribe messaging protocol commonly used for IoT sensor telemetry.
MoSCoW	Prioritisation technique: Must, Should, Could, Won't have.
Zone	A defined geographic sub-area of the monitored forest range used as the unit of risk scoring and alerting.

1.6 References

- Assignment problem statement 30 — "Smart Forest Fire Detection and Environmental Monitoring System" (industry context, problem statement, objectives, expected outcomes).
- CSA1013 Software Engineering — Agile Sprint Planning & User Story Workshop report for this project (Epics, user stories, acceptance criteria, backlog, and sprint plan), used here as the traceability source between backlog items and formal requirements.
- IEEE 830 / ISO/IEC/IEEE 29148 conventions for SRS structuring (used informally as a structural template for Sections 3–9 of this document).

2. Problem Description

2.1 Business Domain

The system operates in the environmental and public-safety domain, specifically wildfire management for government forest departments and allied emergency-response agencies. The domain combines real-time IoT sensing, geospatial data (drone/satellite), AI-based pattern detection, and multi-channel emergency communication — making it simultaneously a monitoring system, a decision-support system, and an alerting system.

2.2 Existing Challenges

- Manual observation and periodic satellite passes detect fires only after significant spread, leaving a narrow and often-missed containment window.
- Environmental data (temperature, humidity, smoke, wind) is not continuously or centrally visible to decision-makers; ground truth is fragmented across sites.
- There is no systematic, explainable way to anticipate where a fire is likely to start next, so patrols and drone coverage are reactive rather than risk-prioritised.
- Remote and mountainous terrain is poorly covered by ground sensors alone, leaving blind spots that only aerial or orbital sensing can close.
- Alerts, when raised, do not reliably or quickly reach every relevant responder, field officer, or nearby community, and unacknowledged critical alerts have no automatic escalation.

- Historical incident and environmental data is not consistently captured in a form usable for policy planning, research, or year-over-year model improvement.

2.3 Stakeholder Analysis

The stakeholder groups below shape requirement priority and later validate that this SRS addresses every stakeholder's concern (see Section 12.1, Traceability Matrix).

Stakeholder	Category	Primary Interest in the System
Forest Department / Government Environmental Agencies	Business Owner	Approves scope, sets compliance and reporting priorities.
Forest Fire Management Authorities	Key Business Stakeholder	Verifies AI-generated alerts; defines detection-accuracy expectations.
Fire & Emergency Response Services	End User	Consumes alerts, location data, and spread/risk information to act.
Forest Rangers / Field Officers	End User	Primary field consumers of mobile alerts and live conditions.
Local & Indigenous Communities	Secondary End User	Receives early warnings; may report incidents from the ground.
Meteorological & Climate Agencies	External Data Provider	Supplies weather / wind / drought-index feeds.
Environmental Research Organisations	Data Consumer	Uses historical exports for ecological research.
System Administrators	Technical Stakeholder	Owns security, access control, uptime, and deployment concerns.

2.4 System Objectives

1. Continuously monitor forest environmental conditions using distributed IoT sensors.
2. Detect forest fires at an early, still-containable stage using AI-based analysis.
3. Predict zone-level fire risk from historical, vegetation, and weather data.
4. Integrate drone and satellite monitoring to extend coverage to remote terrain.
5. Generate low-latency, prioritised emergency alerts to the right responders.
6. Support firefighting operations with spread and resource-deployment context.
7. Produce environmental analytics and reporting for long-term management and policy.
8. Contribute to improved forest conservation and biodiversity protection outcomes.

3. Scope of the System

3.1 In Scope

- Role-based user account management, authentication, and audit logging for all human actors.
- Ingestion and live display of ground IoT sensor telemetry (temperature, humidity, smoke, wind) per zone, including automatic anomaly/offline flagging.
- An AI/computer-vision detection pipeline that fuses sensor and imagery data to raise fire-detection alerts within a defined latency target, plus a human verification workflow.

- A scheduled, explainable zone-level fire-risk scoring engine driven by weather, drought-index, vegetation, and historical data.
- Ingestion of drone-collected imagery and satellite thermal-anomaly feeds, and automatic/on-demand drone patrol scheduling.
- Multi-channel (SMS, email, in-app) alert delivery with configurable thresholds and automatic escalation of unacknowledged critical alerts.
- Scheduled and on-demand analytical, incident, and detection-accuracy reporting, including historical data export.
- A role-appropriate, filterable command dashboard consolidating sensor status, risk heatmap, and active incidents.

3.2 Out of Scope (for this phase / pilot)

- Physical procurement, installation, and field maintenance of IoT hardware, drones, and satellite ground stations (treated as a parallel, non-software workstream).
- Training a fire-detection computer-vision model from scratch — a baseline pre-trained model is assumed and only fine-tuning/retraining on confirmed-incident feedback is in scope.
- Direct dispatch or command of physical firefighting crews and equipment (the system informs and supports responders; it does not control field operations).
- Multi-zone / multi-region scale-out — the initial release targets a single pilot forest range, with scale-out treated as a later architectural milestone.
- Public-facing citizen self-reporting features beyond receiving early-warning notifications (e.g., no citizen-submitted incident-photo pipeline in this phase).

3.3 Product Perspective

The system is a new, self-contained platform rather than a replacement for an existing automated system — it replaces manual observation and infrequent satellite-only monitoring. It integrates with, but does not own, several external systems: meteorological/drought-index data providers, satellite thermal-feed providers, drone flight-control systems, and SMS/email gateways. Internally, it is composed of an IoT ingestion layer, an AI detection and risk-scoring layer, a notification/escalation layer, a reporting layer, and a command dashboard, all sitting behind a common authentication and role-based access-control layer.

4. Actors and Use Cases

4.1 Primary Actors

Actor	Nature	Description
System Administrator	Human, Primary	Manages accounts, roles, sensor health, and system configuration.
Forest Department Official	Human, Primary	Views monitoring data, risk scores, dashboards, and reports for planning and policy.
Forest Fire Management Authority	Human, Primary	Verifies/dismisses AI alerts; requests drone flights; owns detection-accuracy oversight.
Field Ranger / Forest Officer	Human, Primary	Ground-level consumer of live conditions, threshold warnings, and mobile alerts.

Actor	Nature	Description
Emergency Response Coordinator	Human, Primary	Coordinates response using the dashboard, alerts, and escalation workflow.

4.2 Secondary Actors

Actor	Nature	Description
Local / Indigenous Community Member	Human, Secondary	Receives early-warning notifications for their area.
Meteorological & Climate Agency	External System, Secondary	Supplies weather and drought-index data consumed by risk scoring.
Satellite / Drone Data Provider	External System, Secondary	Supplies thermal-anomaly and aerial imagery feeds.
Environmental Research Organisation	Human/System, Secondary	Consumes historical exports for research; not an operational actor.

4.3 Use Case Summary

The table below lists the major use cases grouped by the actor who primarily initiates them; several use cases are shared across actors (e.g., login) and several system-facing use cases are triggered by scheduled jobs rather than a human actor, shown as "System (scheduled)".

Use Case ID	Use Case	Primary Actor(s)
UC-1	Manage user accounts and roles	System Administrator
UC-2	Log in securely (role-based)	All human actors
UC-3	Reset password via verified channel	All human actors
UC-4	Monitor real-time environmental data	Forest Department Official, Field Ranger
UC-5	Flag anomalous / offline sensors	System (scheduled); System Administrator
UC-6	Receive in-app threshold warning	Field Ranger
UC-7	Detect fire signature (AI/CV)	System (scheduled); Emergency Response Coordinator (consumer)
UC-8	Verify or dismiss a fire-detection alert	Forest Fire Management Authority
UC-9	View zone fire-risk score	Forest Department Official, Forest Fire Management Authority
UC-10	Ingest weather / drought-index data	Meteorological & Climate Agency (system)
UC-11	Schedule / dispatch drone patrol	System (scheduled); Forest Fire Management Authority
UC-12	Request on-demand drone flight	Forest Fire Management Authority
UC-13	Ingest satellite thermal feed	Satellite / Drone Data Provider (system)
UC-14	Receive multi-channel alert	Field Ranger, Emergency Response Coordinator, Community Member

Use Case ID	Use Case	Primary Actor(s)
UC-15	Escalate unacknowledged alert	System (scheduled); Emergency Response Coordinator
UC-16	Generate analytical / accuracy reports	Forest Department Official, Environmental Research Organisation
UC-17	View live command dashboard	Emergency Response Coordinator, Forest Department Official, Field Ranger
UC-18	Filter dashboard by zone / risk / status	Emergency Response Coordinator

4.4 Use Case Diagram

The diagram below shows the system boundary, all primary and secondary actors, and their association with the eighteen major use cases identified above.

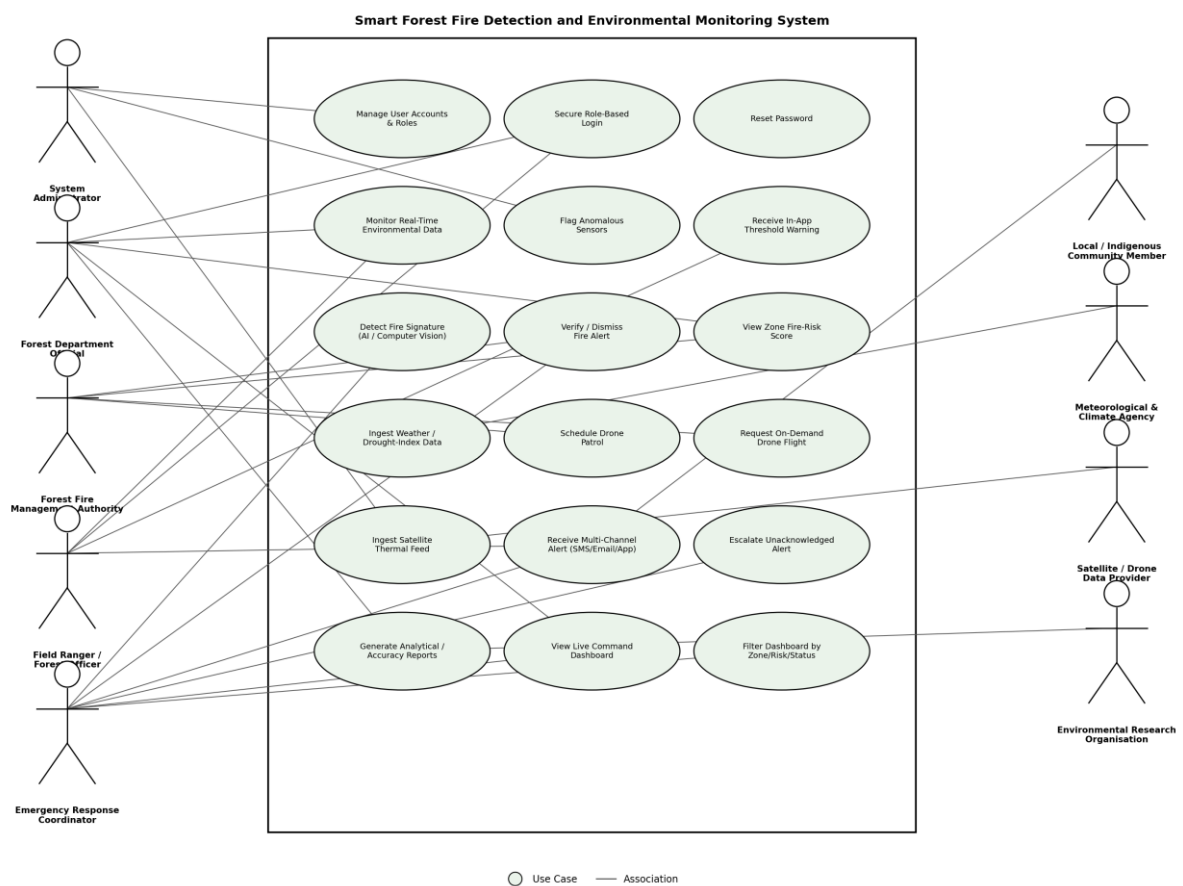


Figure 4.1 — Use case diagram for the AI-Powered Smart Forest Fire Detection and Environmental Monitoring System.

5. Functional Requirements

Functional requirements are grouped by the eight system modules identified during scenario analysis. Priority follows MoSCoW and is carried over directly from the project's backlog-prioritisation exercise so that this SRS remains consistent with the sprint plan.

5.1 Module 1 — User Management & Access Control

ID	Requirement	Priority
FR-1.1	The system shall allow a System Administrator to create role-based user accounts (Administrator, Forest Department Official, Fire Management Authority, Field Ranger, Emergency Coordinator, Community Member).	Must
FR-1.2	The system shall authenticate users with hashed credentials and redirect each user to the dashboard view configured for their role.	Must
FR-1.3	The system shall lock an account for a cooldown period after three consecutive failed login attempts and record an audit-log entry.	Must
FR-1.4	The system shall allow a System Administrator to deactivate or reassign a user's role.	Should
FR-1.5	The system shall record every login attempt (success and failure) in an immutable audit log.	Must
FR-1.6	The system shall allow a user to reset their password via a time-limited link or code sent to a verified email or phone number.	Should

5.2 Module 2 — Environmental Monitoring

ID	Requirement	Priority
FR-2.1	The system shall ingest temperature, humidity, smoke, and wind readings from IoT sensors and display updated values for a zone within 5 seconds of transmission.	Must
FR-2.2	The system shall mark a sensor "Offline" and flag it for maintenance if it does not transmit for more than 10 minutes.	Should
FR-2.3	The system shall flag a sensor "Anomalous" and create a maintenance ticket after three consecutive out-of-range readings.	Should
FR-2.4	The system shall display historical environmental trends for a zone over a user-selected date range.	Should
FR-2.5	The system shall push an in-app warning to Field Rangers, distinct in styling from a confirmed fire alert, within 30 seconds of a zone's readings crossing a configured danger threshold.	Should

5.3 Module 3 — Fire Detection (AI / Computer Vision)

ID	Requirement	Priority
FR-3.1	The system shall fuse sensor and imagery data and generate a fire-detection alert within 60 seconds when AI model confidence for a fire signature exceeds a configured threshold.	Must
FR-3.2	The system shall allow a Forest Fire Management Authority user to mark an alert "Verified" or "False Positive" with a single action.	Must
FR-3.3	The system shall persist verification outcomes as labelled feedback for scheduled model retraining.	Should

ID	Requirement	Priority
FR-3.4	The system shall display the model's confidence score and the specific sensor readings or imagery frames that triggered each alert.	Should

5.4 Module 4 — Fire Risk Prediction

ID	Requirement	Priority
FR-4.1	The system shall run a scheduled job that publishes a zone-level fire-risk score (Low / Medium / High / Extreme) with a timestamp, combining weather, vegetation-dryness, and historical fire data.	Must
FR-4.2	The system shall automatically ingest weather and drought-index feeds from the meteorological provider on a scheduled sync, making values available to risk scoring within 15 minutes.	Should
FR-4.3	If the meteorological feed is unavailable, the system shall use the last known values and flag the affected risk score as "stale data."	Should
FR-4.4	The system shall display the top three contributing factors behind any published risk score.	Should
FR-4.5	The system shall provide a ranked, cross-zone view of current risk scores in descending order.	Could

5.5 Module 5 — Drone & Satellite Integration

ID	Requirement	Priority
FR-5.1	The system shall automatically queue a drone patrol for any zone whose risk score changes to High or Extreme.	Should
FR-5.2	The system shall ingest satellite thermal-anomaly data and geo-align it with the corresponding zone boundary within one polling cycle.	Should
FR-5.3	The system shall display drone imagery, satellite thermal data, and ground-sensor alerts fused on a single map view.	Should
FR-5.4	The system shall allow an authorised Forest Fire Management Authority user to submit coordinates for an on-demand drone flight and notify the requester once imagery is available.	Could

5.6 Module 6 — Reporting

ID	Requirement	Priority
FR-6.1	The system shall generate scheduled daily, weekly, and seasonal reports covering environmental trends, incidents, and detection accuracy, available for download.	Could
FR-6.2	The system shall allow an Environmental Research Organisation user to export historical sensor and incident data.	Could
FR-6.3	The system shall generate a detection-accuracy report comparing verified alerts to false positives and trend this ratio against the previous reporting period.	Could

5.7 Module 7 — Notification & Alerting

ID	Requirement	Priority
FR-7.1	The system shall dispatch SMS, email, and in-app notifications to all subscribed users for a zone within 60 seconds of a confirmed fire-detection alert.	Must
FR-7.2	The system shall allow a System Administrator to configure alert thresholds per zone, taking effect immediately on save.	Should
FR-7.3	The system shall log a delivery-configuration warning if a user with no configured notification channel is targeted by an alert.	Should
FR-7.4	The system shall automatically escalate a critical alert to the next responder tier, and log the escalation, if it is not acknowledged within 5 minutes.	Should

5.8 Module 8 — Command Dashboard

ID	Requirement	Priority
FR-8.1	The system shall load live sensor status, the fire-risk heatmap, and active incidents on the Command Dashboard within 2 seconds.	Must
FR-8.2	The system shall show only zones and widgets relevant to the logged-in user's role, excluding administrative controls for non-administrative roles.	Should
FR-8.3	The system shall surface a newly issued alert for a Field Ranger's assigned zone at the top of their alert feed within 2 seconds of generation.	Must
FR-8.4	The system shall allow the dashboard's map, heatmap, and incident list to be filtered by zone, risk level, or incident status without a full page reload.	Could

Total functional requirements identified: 32, spanning all eight modules. This count and grouping is directly traceable to the 23 user stories and their acceptance criteria produced during backlog refinement, with several user stories decomposed into more than one atomic functional requirement for specification clarity.

6. Non-Functional Requirements

Non-functional requirements are grouped under the categories specified for this analysis: performance, security, reliability, usability, scalability, maintainability, and availability, with a data-retention category added given the platform's reporting and research-export obligations.

6.1 Performance

ID	Category	Requirement
NFR-1	Performance	A fire-detection alert shall be generated within 60 seconds of a confirmed anomaly (95th percentile).
NFR-2	Performance	The Command Dashboard shall load within 2 seconds on both desktop and mobile devices under normal load.
NFR-3	Performance	Live environmental readings shall propagate to the dashboard within 5 seconds of sensor transmission.
NFR-4	Performance	Dashboard filter operations shall update the map/heatmap/incident list without a full page reload and within 1 second.

6.2 Security

ID	Category	Requirement
NFR-5	Security	All user passwords shall be stored using a salted, one-way hashing algorithm; plaintext passwords shall never be persisted or logged.
NFR-6	Security	Access to zone data, alerts, and reports shall be enforced by role-based access control at the API layer, not only in the UI.
NFR-7	Security	All data in transit (sensor telemetry, dashboard traffic, external feed sync) shall be encrypted using TLS 1.2 or higher.
NFR-8	Security	Every authentication event and every alert-verification action shall be attributable to a specific user in an immutable audit log.

6.3 Reliability

ID	Category	Requirement
NFR-9	Reliability	The platform shall maintain 99.5% uptime for the ingestion, detection, and alerting path, measured monthly.
NFR-10	Reliability	If the meteorological or satellite feed is unavailable, the risk-scoring job shall degrade gracefully using last-known data rather than failing silently.
NFR-11	Reliability	Notification dispatch shall implement per-channel retry logic so a transient SMS/email gateway failure does not silently drop an alert.

6.4 Usability

ID	Category	Requirement
NFR-12	Usability	Field-facing views (mobile alerts, threshold warnings) shall be usable on low-end Android devices and remain legible with limited connectivity.
NFR-13	Usability	A confirmed fire alert shall be visually and audibly distinguishable from an informational threshold warning at first glance.
NFR-14	Usability	Dashboard widgets shown to a given role shall exclude controls that role is not authorised to act on, to avoid confusing or unsafe actions.

6.5 Scalability

ID	Category	Requirement
NFR-15	Scalability	The ingestion architecture shall support scale-out from a single pilot forest range to multiple zones/regions without a redesign of the sensor-data schema.
NFR-16	Scalability	The notification service shall support an increasing subscriber base per zone without a proportional increase in per-alert delivery latency.

6.6 Maintainability

ID	Category	Requirement
NFR-17	Maintainability	The AI detection model shall be retrainable on newly confirmed-incident feedback without requiring a redeployment of the ingestion or alerting services.
NFR-18	Maintainability	API contracts, data schemas, and deployment notes shall be kept current as part of each sprint's Definition of Done.

6.7 Availability

ID	Category	Requirement
NFR-19	Availability	The alerting and dashboard services shall remain available during planned maintenance of non-critical modules (e.g., reporting) via independent deployability.
NFR-20	Availability	Field Rangers shall be able to view the most recently cached readings for their zone even during a brief loss of mobile connectivity.

6.8 Data Retention

ID	Category	Requirement
NFR-21	Data Retention	Sensor, alert, and incident data shall be retained for a minimum of 5 years to support research export and seasonal trend reporting.
NFR-22	Data Retention	Exported historical data shall be provided in a non-proprietary, machine-readable format (e.g., CSV/JSON) for research consumers.

7. Use Case Descriptions

Full "fully-dressed" descriptions are provided below for the six use cases that anchor the platform's core value chain — sense, detect, verify, score, notify, and observe. The remaining twelve use cases listed in Section 4.3 follow the same template and are summarised at requirement level in Section 5.

UC-2 — Log in securely (role-based)

Field	Description
Primary Actor	Any registered human actor
Preconditions	The user holds an active, non-locked account with an assigned role.
Postconditions	The user is authenticated and viewing the dashboard configured for their role; a login audit entry exists.

Main Flow:

9. The user opens the login screen and enters their username and password.
10. The system validates the credentials against stored, hashed values.
11. On success, the system creates an audit-log entry and redirects the user to the dashboard view configured for their role.

Alternate Flow:

- The user selects "Forgot password" and is routed to UC-3 (Reset Password).

Exceptions:

- On a fourth consecutive failed attempt within 15 minutes, the account is temporarily locked and an audit-log entry is created (FR-1.3).

UC-7 — Detect fire signature (AI / Computer Vision)

Field	Description
Primary Actor	System (scheduled ingestion/inference job); consumed by Emergency Response Coordinator
Preconditions	Fused sensor and imagery data for a zone is available to the inference pipeline; a baseline detection model is deployed.
Postconditions	A fire-detection alert exists with a confidence score and linked evidence, or no alert is raised if confidence is below threshold.

Main Flow:

12. The ingestion pipeline receives new sensor and/or imagery data for a zone.
13. The AI model scores the fused input for fire-signature confidence.
14. If confidence exceeds the configured threshold, the system generates and logs a fire-detection alert within 60 seconds of the confirmed anomaly.
15. The alert is surfaced to subscribed responders (see UC-14) and to the Command Dashboard (UC-17).

Alternate Flow:

- If the imagery feed is temporarily unavailable, the model scores on sensor data alone with a reduced-confidence flag.

Exceptions:

- If model inference exceeds the latency budget, the event is logged for the reliability NFR (NFR-1) and a degraded-mode alert may be raised on sensor thresholds alone.

UC-8 — Verify or dismiss a fire-detection alert

Field	Description
Primary Actor	Forest Fire Management Authority
Preconditions	An active, unverified fire-detection alert exists (from UC-7).
Postconditions	The alert is marked "Verified" or "False Positive"; the outcome is stored as labelled feedback for model retraining.

Main Flow:

- 16.The authority user opens the alert detail view, including confidence score and supporting evidence (FR-3.4).
- 17.The user selects "Verified" or "False Positive" as a single action.
- 18.The system persists the outcome and timestamps the decision.
- 19.The outcome is added to the feedback dataset used in the next model-retraining cycle.

Alternate Flow:

- The user may add an optional note explaining the dismissal, aiding future model-error analysis.

Exceptions:

- If the user closes the view without deciding, the alert remains "Unverified" and continues to appear in the pending-verification queue.

UC-9 — View zone fire-risk score

Field	Description
Primary Actor	Forest Department Official; Forest Fire Management Authority
Preconditions	The scheduled risk-scoring job (FR-4.1) has published at least one score for the zone.
Postconditions	The user has viewed the current score, its category, and its top contributing factors.

Main Flow:

- 20.The user opens a zone's detail view.
- 21.The system displays the current risk score (Low/Medium/High/Extreme) with its timestamp.
- 22.The system displays the top three contributing factors behind the score (FR-4.4).

Alternate Flow:

- The user opens the cross-zone ranking view (UC not separately detailed) to compare risk across all zones (FR-4.5).

Exceptions:

- If the meteorological feed was unavailable at scoring time, the displayed score is flagged "stale data" (FR-4.3).

UC-14 — Receive multi-channel alert

Field	Description
Primary Actor	Field Ranger; Emergency Response Coordinator; Local Community Member
Preconditions	The recipient is subscribed to notifications for the affected zone and has at least one configured channel.
Postconditions	The recipient has received the alert on all configured channels (SMS, email, in-app) within 60 seconds of alert generation.

Main Flow:

- 23.A fire-detection alert is confirmed (from UC-7/UC-8).
- 24.The notification service resolves all subscribers for the affected zone.
- 25.The system dispatches SMS, email, and in-app notifications to each subscriber within 60 seconds.
- 26.Delivery status per channel is recorded.

Alternate Flow:

- A Field Ranger with intermittent connectivity receives the SMS channel first and the in-app notification on next sync.

Exceptions:

- If a subscriber has no configured channel, the system logs a delivery-configuration warning for the administrator instead of failing silently (FR-7.3).

UC-17 — View live command dashboard

Field	Description
Primary Actor	Emergency Response Coordinator; Forest Department Official; Field Ranger
Preconditions	The user is authenticated (UC-2) and has dashboard access appropriate to their role.
Postconditions	The dashboard is displaying live, role-appropriate data within the performance target.

Main Flow:

- 27.The authenticated user opens the Command Dashboard.
- 28.The system loads live sensor status, the fire-risk heatmap, and active incidents within 2 seconds.
- 29.The system restricts widgets and controls to those appropriate for the user's role (FR-8.2).
- 30.New alerts for the user's assigned zone(s) appear at the top of their feed within 2 seconds of generation.

Alternate Flow:

- The user applies a zone, risk-level, or incident-status filter (UC-18); the view updates without a full page reload.

Exceptions:

- If live data cannot be loaded within the performance target, the dashboard shows the most recently cached state with a visible "stale" indicator rather than a blank screen.

8. Data Requirements

8.1 Core Data Entities

Entity	Description	Key Attributes
User	A human actor with system access.	user_id, name, role, contact_channel(s), status, created_at
Role	A named access-control profile.	role_id, role_name, permission_set
Zone	A geographic sub-area of the monitored forest range.	zone_id, name, boundary_geometry, assigned_rangers
Sensor	A physical IoT device reporting environmental readings.	sensor_id, zone_id, type, status (Active/Offline/Anomalous), last_seen
SensorReading	A single telemetry data point.	reading_id, sensor_id, timestamp, temperature, humidity, smoke_level, wind_speed
FireAlert	An AI-generated or escalated fire-detection event.	alert_id, zone_id, timestamp, confidence_score, evidence_ref, status (Unverified/Verified/False Positive), verified_by
RiskScore	A scheduled zone-level risk publication.	score_id, zone_id, timestamp, category, contributing_factors, data_freshness_flag
WeatherFeedRecord	An ingested meteorological/drought-index data point.	record_id, zone_id, timestamp, source, values
SatelliteFeedRecord	An ingested thermal-anomaly reading.	record_id, zone_id, timestamp, thermal_signature, geo_alignment
DroneRequest	A scheduled or on-demand drone-flight task.	request_id, zone_id/coordinates, requested_by, status, imagery_ref
Notification	A dispatched alert delivery attempt.	notification_id, alert_id, recipient_id, channel, delivery_status, sent_at
Escalation	A record of an unacknowledged-alert escalation.	escalation_id, alert_id, escalated_to, escalated_at
Report	A generated analytical or accuracy report.	report_id, type, period, generated_at, download_ref
AuditLogEntry	An immutable record of a security-relevant action.	log_id, user_id, action, timestamp, outcome

8.2 Data Relationships (Narrative)

- A Zone has many Sensors; a Sensor produces many SensorReadings over time.
- A FireAlert belongs to exactly one Zone and is derived from a window of SensorReadings and/or imagery evidence; it is optionally linked to a verifying User.
- A RiskScore belongs to exactly one Zone and references the WeatherFeedRecord(s) and historical data used to compute it.
- A Notification always references the FireAlert (or threshold event) that triggered it, and the recipient User; an Escalation references the Notification/FireAlert chain it grew out of.

- A DroneRequest may be system-generated (linked to a RiskScore transition) or user-generated (linked to the requesting User) and, once fulfilled, links to the resulting SatelliteFeedRecord/imagery evidence.

8.3 Data Volume, Retention, and Quality

- Sensor telemetry is expected at sub-minute frequency per sensor; the ingestion path must sustain this volume per zone without degrading dashboard latency (NFR-1, NFR-2).
- Historical sensor, alert, and incident data is retained for a minimum of five years (NFR-21) to support seasonal trend reporting and research export.
- Data quality rules: readings outside a sensor's calibrated physical range for three consecutive samples trigger the "Anomalous" flag (FR-2.3); a risk score computed on out-of-date external data is explicitly flagged "stale" rather than presented as current (FR-4.3).

9. External Interface Requirements

9.1 User Interfaces

- A responsive web dashboard for Forest Department Officials, Fire Management Authorities, and Emergency Response Coordinators, providing the Command Dashboard, risk views, alert verification, and reporting.
- A mobile-optimised interface (native app or responsive web) for Field Rangers, prioritising fast load times, offline-tolerant cached views, and clear visual/audible differentiation between informational warnings and confirmed alerts (NFR-12, NFR-13).
- An administrative console for System Administrators covering account/role management, sensor health, and alert-threshold configuration.

9.2 Hardware Interfaces

- IoT ground sensors (temperature, humidity, smoke, wind) communicating over a lightweight publish-subscribe protocol (e.g., MQTT) to the ingestion service.
- Drone flight-control hardware/systems that accept scheduled or on-demand flight tasks and return captured imagery.

9.3 Software Interfaces

Interface	Direction	Purpose
Meteorological / Drought-Index Provider API	Inbound	Scheduled sync of weather and drought-index values feeding FR-4.1/FR-4.2.
Satellite Thermal-Feed Provider API	Inbound	Ingestion of thermal-anomaly data feeding FR-5.2.
SMS Gateway	Outbound	Delivery of SMS alerts and escalations (FR-7.1, FR-7.4).
Email Gateway (SMTP/API)	Outbound	Delivery of email alerts, reports, and password-reset messages.
AI Model Serving Endpoint	Internal	Hosts the fire-detection and (future) risk-contributing-factor models.
Reporting / Export Service	Outbound	Produces downloadable report files and research data exports (FR-6.1–FR-6.3).

9.4 Communication Interfaces

- All external and internal API traffic shall use HTTPS/TLS (NFR-7).
- Sensor-to-ingestion telemetry shall use a lightweight, low-bandwidth protocol suited to constrained field connectivity.
- Real-time dashboard updates shall use a push mechanism (e.g., WebSocket or server-sent events) to meet the 2-second alert-surfacing target (FR-8.3) without client polling.

10. System Features

System features summarise the platform's major delivered capabilities at a level a Product Owner or evaluator can scan without reading every individual functional requirement, with priority and traceability back to Section 5.

Feature	Priority	Description	Related FRs
Secure Role-Based Access	Must	Account creation, authentication, lockout, audit logging, and password reset.	FR-1.1–FR-1.6
Live Environmental Monitoring	Must	Real-time zone readings with automatic offline/anomalous sensor flagging and historical trends.	FR-2.1–FR-2.5
AI Fire Detection & Verification	Must	Sub-60-second AI alerting from fused sensor/imagery data, with human verification and retraining feedback.	FR-3.1–FR-3.4
Explainable Fire-Risk Scoring	Must	Scheduled, explainable zone risk scores driven by weather and historical data.	FR-4.1–FR-4.5
Drone & Satellite Coverage Extension	Should	Automatic and on-demand drone scheduling plus satellite thermal-feed fusion.	FR-5.1–FR-5.4
Analytics & Compliance Reporting	Could	Scheduled and on-demand reports, plus historical research export.	FR-6.1–FR-6.3
Multi-Channel Alerting & Escalation	Must	SMS/email/in-app delivery with configurable thresholds and 5-minute escalation.	FR-7.1–FR-7.4
Unified Command Dashboard	Must	Role-appropriate live map, heatmap, incident list, and filtering.	FR-8.1–FR-8.4

11. Assumptions and Constraints

11.1 Assumptions

- The pilot deployment targets a single forest range before any multi-zone scale-out is attempted.
- IoT hardware procurement and field installation proceed as a parallel, non-software workstream; software components assume sensor connectivity is available in time for integration testing.
- A baseline, pre-trained computer-vision model is available for initial integration; training a detection model from scratch is out of scope and would materially change effort estimates.
- Meteorological and satellite data providers expose stable, documented APIs; any provider-side format change is treated as an external risk, not a defect in this system.
- End users (Field Rangers, community members) have at least intermittent mobile connectivity sufficient for SMS or periodic app sync, even if not constant broadband access.

11.2 Constraints

Type	Constraint
Technical	The detection pipeline must meet a 60-second alert-generation target and the dashboard a 2-second load target; both are hard performance constraints on architecture choices (e.g., streaming ingestion, push-based dashboard updates).

Type	Constraint
Technical	Sensor telemetry must be ingested over a low-bandwidth-friendly protocol suited to field connectivity conditions.
Operational	External dependencies (meteorological feed, satellite feed) are outside the project team's direct control and must degrade gracefully rather than block core detection/alerting.
Operational	IoT hardware installation timelines are managed by a separate workstream and may not align perfectly with the software sprint schedule.
Economic	The pilot scope is bounded to a single forest range partly to control cost before committing to multi-zone hardware and licensing spend.
Legal / Compliance	Location and personal contact data (for alert delivery) must be access-controlled and audit-logged given its sensitivity to Indigenous and local communities and field personnel safety.
Time	A working sense-detect-verify-alert loop is targeted by the end of the second sprint, ahead of the wider pilot go-live milestone.

12. Requirement Validation — Completeness and Consistency

12.1 Traceability: Stakeholder Concern → Requirement

Each stakeholder group identified in Section 2.3 is mapped below to the requirements that address its primary concern, confirming no stakeholder group is left without at least one directly traceable requirement.

Stakeholder	Primary Concern	Addressed By
Forest Department / Government Agencies	Compliance, coverage, reporting, scope approval	FR-6.1–FR-6.3, FR-8.2, NFR-21, NFR-22
Forest Fire Management Authorities	Alert accuracy, verification, drone tasking	FR-3.1–FR-3.4, FR-5.4, FR-6.3
Fire & Emergency Response Services	Timely, prioritised alerts and situational awareness	FR-7.1, FR-7.4, FR-8.1, FR-8.3, FR-8.4
Forest Rangers / Field Officers	Ground conditions, mobile alerts, usability	FR-2.1, FR-2.5, FR-7.1, NFR-12, NFR-13
Local & Indigenous Communities	Early warning, privacy of contact/location data	FR-7.1, FR-7.3, NFR-6, NFR-8
Meteorological & Climate Agencies	Reliable, well-specified data ingestion contract	FR-4.2, FR-4.3, Section 9.3
Environmental Research Organisations	Historical, exportable data	FR-6.2, NFR-21, NFR-22
System Administrators	Security, uptime, access control	FR-1.1–FR-1.5, NFR-5–NFR-8, NFR-9

12.2 Completeness Check

- Every one of the 8 system objectives in Section 2.4 is covered by at least one functional-requirement module (mapping: monitoring → Module 2; detection → Module 3; prediction → Module 4; drone/satellite → Module 5; alerts → Module 7; firefighting support → Modules 4/5/8; analytics → Module 6; conservation outcome → cumulative effect of all modules).
- Every actor identified in Section 4.1/4.2 initiates or receives at least one use case, and every use case maps to at least one functional requirement.
- Every non-functional category requested for this assignment (performance, security, reliability, usability, scalability, maintainability, availability) has at least two concrete, testable requirements rather than a generic statement.

12.3 Ambiguity, Redundancy, and Inconsistency Review

- Ambiguity resolved: the original problem statement's phrase "predict fire risks" was ambiguous between a binary flag and a graded score; this SRS resolves it as an explicit four-level category (Low/Medium/High/Extreme) with named contributing factors (FR-4.1, FR-4.4), matching the acceptance-criteria precedent in the referenced sprint-planning workshop.
- Ambiguity resolved: "immediate alerts" in the source problem statement is quantified as a 60-second generation target (FR-3.1) and a 60-second multi-channel dispatch target (FR-7.1), so both are independently testable rather than left as a subjective adjective.
- Potential redundancy checked: FR-2.5 (in-app threshold warning) and FR-7.1 (multi-channel confirmed alert) were reviewed to confirm they are distinct events, not a duplicated requirement — the former is a pre-alert informational warning at the sensor-threshold level, the latter is the confirmed, AI-verified fire

alert; NFR-13 makes the visual/audible distinction an explicit requirement precisely because the two are easy to conflate.

- Inconsistency checked: drone-patrol scheduling appears both as an automatic, risk-triggered behaviour (FR-5.1) and a manual, on-demand action (FR-5.4). These were confirmed as two independent triggers of the same DroneRequest entity (Section 8.1) rather than a conflicting pair of requirements.
- Missing-requirement gap closed: the original problem statement did not explicitly mention account security or audit logging; Module 1 (FR-1.1–FR-1.6) and NFR-5–NFR-8 were added because every other feature depends on knowing who is authorised to view or act on sensitive location and alert data.

12.4 Suggested Improvements

31. Add an explicit non-functional requirement for AI model accuracy/precision-recall targets once the first season of verified/false-positive data (FR-3.3) is available, since "detect early" currently has a latency target but not yet a formally agreed accuracy target.
32. Introduce a formal data-privacy requirement for community-member contact information (opt-in/opt-out consent, retention limits distinct from the 5-year operational retention in NFR-21), given the sensitivity of alerting an Indigenous or local community's personal contact channels.
33. Add a defined non-functional requirement for sensor time-synchronisation (clock drift tolerance) since risk scoring and fire detection both depend on timestamp accuracy across a distributed sensor network.
34. Consider a formal capacity/NFR statement for concurrent dashboard users during a live multi-zone incident, since current dashboard performance targets (NFR-2) are specified per-request rather than under peak concurrent load.

13. Summary and Next Steps

This SRS translates the AI-Powered Smart Forest Fire Detection and Environmental Monitoring System problem statement into a complete, traceable requirement specification: a stakeholder and problem analysis (Sections 1–2), a bounded scope statement (Section 3), a full actor and use-case model with a use-case diagram (Section 4), 32 functional requirements across eight modules (Section 5), 22 non-functional requirements across seven categories (Section 6), six fully-dressed use-case descriptions (Section 7), a core data model (Section 8), external interface requirements (Section 9), a system-feature summary (Section 10), assumptions and constraints (Section 11), and a closing validation pass confirming completeness and resolving ambiguity, redundancy, and inconsistency (Section 12).

The immediate next step is stakeholder and Product Owner sign-off on this SRS, followed by mapping each functional requirement to the corresponding backlog items and sprint plan already produced for this project, so implementation can proceed directly from an agreed, unambiguous specification rather than from the informally stated problem statement alone.

14. Glossary

Term	Definition
Actor	A person or external system that interacts with the system to achieve a goal.
Use Case	A description of a sequence of actions a system performs to yield an observable result for an actor.

Term	Definition
Functional Requirement	A statement of a specific behaviour or function the system must provide.
Non-Functional Requirement	A quality attribute or constraint on how the system performs its functions (performance, security, etc.).
Actor Association	The relationship line in a use-case diagram connecting an actor to a use case it participates in.
System Boundary	The rectangle in a use-case diagram delimiting what is inside versus outside the system under specification.
Traceability	The ability to follow a requirement back to its originating stakeholder need, and forward to its implementation and test.
MoSCoW	A prioritisation technique categorising requirements as Must, Should, Could, or Won't have.
Stale Data Flag	An explicit indicator that a displayed value (e.g., a risk score) was computed from out-of-date external data.